Project Intel V2

Technical Design Document

Version 0.3 (draft)

Last Updated: February 28, 2026

Author: Alan

1. Overview

Project Intel V2 is a complete rewrite of the V1 CLI tool, now as a desktop application using Tauri framework. The goal is to provide an AI-powered project management assistant that runs entirely locally for GDPR compliance.

2. Architecture

The system consists of three main components:

- • Tauri Desktop App (Rust + React frontend)
- • FastAPI Backend (Python 3.11+)
- • Ollama LLM Service (local inference)

Wait should this be 4 components if we count SQLite separately? Anyway...

3. Database Schema

We settled on 8 tables after much deliberation. Original design had 6 but added scope_items and dependencies after realizing those were critical.

Tables:

- - documents: stores uploaded files and extracted text
- - document_types: configurable extraction strategies
- - actions: tasks with owners/due dates/status
- - risks: project risks with impact/likelihood
- - deadlines: critical dates
- - dependencies: task relationships (blocks/enables/relates)
- - scope_items: track scope changes
- - notifications: alerts and briefings

4. LLM Strategy

After testing multiple models, we converged on a three-model approach:

• mistral-nemo (12B): Fast structured extraction

• llama3.1 (8B): General Q&A and summaries

• deepseek-r1 (7B): Deep reasoning and scope analysis

IMPORTANT DECISION: Model selection happens at document type level, not globally. This lets us optimize extraction quality vs speed per document type.

5. Security & Privacy

All processing is local. NO external API calls except during development. CSP restricts frontend to localhost:8000. CORS limits methods to GET/POST/PATCH/DELETE.

RISK: Developer might forget and add OpenAI call - need code review checklist

ACTION: Add pre-commit hook to scan for external API URLs - Alan - March 12

6. Known Issues

• Drag-drop had missing onDragEnter (FIXED)

• Settings router import collision with config (FIXED with alias)

• Excel multi-sheet handling punted to V3, only reading first sheet

• No auto-detection of document types, user must declare at upload

• Pagination hardcoded at 50 items (should be configurable?)

7. Timeline

Original estimate: 6 weeks

Actual timeline: TBD (already week 3, maybe 8-10 weeks?)

DEADLINE: Working prototype demo March 28

DEADLINE: V2.0 release target April 15 2026

TODO: Add diagrams of data flow

TODO: Document all API endpoints

TODO: Actually finish writing this document lol